

Calculus and Linear Algebra

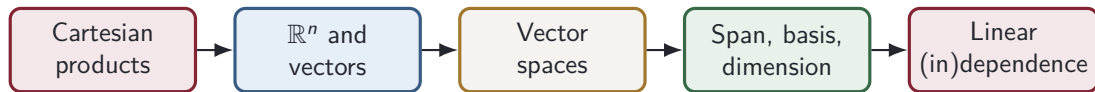
Cartesian Products and Vector Spaces

Lecture 3

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Lecture outline



Aim

To pass from the one-dimensional real line to multidimensional spaces and to introduce the algebraic structure used throughout linear algebra.

Ordered pairs

For sets A and B , an **ordered pair** is written

$$(a, b), \quad a \in A, \quad b \in B.$$

The order is part of the definition:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

$$a \in A$$

$$b \in B$$

$$(a, b)$$

Cartesian product: definition

Definition

The **Cartesian product** of two sets A and B is

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

Example:

$$A = \{1, 2\}, \quad B = \{x, y, z\}.$$

Then

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

A finite Cartesian product

	x	y	z
1	•	•	•
2	•	•	•

$$A = \{1, 2\}$$

$$B = \{x, y, z\}$$

$$|A \times B| = 2 \cdot 3 = 6$$

The order of the factors matters

In general,

$$A \times B \neq B \times A.$$

For

$$A = \{1, 2\}, \quad B = \{x, y\},$$

we have

$$A \times B = \{(1, x), (1, y), (2, x), (2, y)\},$$

while

$$B \times A = \{(x, 1), (x, 2), (y, 1), (y, 2)\}.$$

The first coordinate belongs to the first factor.

Cartesian powers of number sets

For a set A ,

$$A^2 = A \times A, \quad A^3 = A \times A \times A.$$

Hence

$$\mathbb{N}^2, \quad \mathbb{Z}^2, \quad \mathbb{Q}^2, \quad \mathbb{R}^2$$

are sets of ordered pairs, and

$$\mathbb{N}^3, \quad \mathbb{Z}^3, \quad \mathbb{Q}^3, \quad \mathbb{R}^3$$

are sets of ordered triples. Mixed products are also allowed, for example

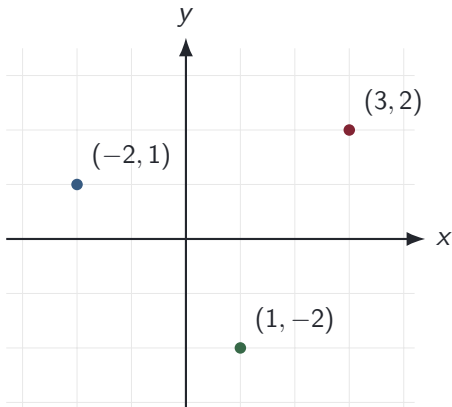
$$\mathbb{Z} \times \mathbb{R} = \{(m, x) : m \in \mathbb{Z}, x \in \mathbb{R}\}.$$

$\mathbb{R}^2$ as the Cartesian plane

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x, y \in \mathbb{R}\}.$$

Each ordered pair determines one point of the plane.

$(3, 2)$, $(-2, 1)$, $(1, -2)$.



From points to vectors in $\mathbb{R}^2$

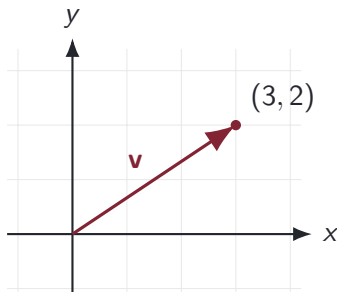
The same pair of real numbers can be written as a column vector:

$$(x, y) \longleftrightarrow \begin{pmatrix} x \\ y \end{pmatrix}.$$

Example:

$$(3, 2) \longleftrightarrow \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

The coordinates specify the displacement from the origin.



$\mathbb{R}^3$: ordered triples

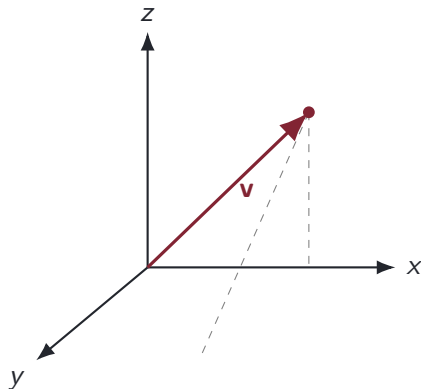
$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}.$$

A vector in $\mathbb{R}^3$ is written

$$\mathbf{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Example:

$$\mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$



$\mathbb{R}^n$: multidimensional vectors

For any positive integer n ,

$$\mathbb{R}^n = \underbrace{\mathbb{R} \times \cdots \times \mathbb{R}}_{n \text{ times}}.$$

An element of $\mathbb{R}^n$ is an n -dimensional vector:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad x_i \in \mathbb{R}.$$

Examples:

$$\begin{pmatrix} 1 \\ 0 \\ -2 \\ 5 \end{pmatrix} \in \mathbb{R}^4, \quad (1, 2, 3, 4, 5) \in \mathbb{R}^5.$$

Exercise: Cartesian products

Exercise

1. Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Write $A \times B$ and $B \times A$.
2. Decide whether each statement is true:

$$(2, a) \in A \times B, \quad (a, 2) \in A \times B.$$

3. Classify each object:

$$(3, -1), \quad \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}, \quad (1, \sqrt{2}), \quad (4, 7, 9, 0).$$

Specify one natural ambient set among $\mathbb{Z}^n, \mathbb{Q}^n, \mathbb{R}^n$.

4. Describe geometrically $\mathbb{R} \times \{0\}$ as a subset of $\mathbb{R}^2$.

Solution: Cartesian products

Solution

1.

$$A \times B = \{(0, a), (0, b), (1, a), (1, b), (2, a), (2, b)\},$$

$$B \times A = \{(a, 0), (a, 1), (a, 2), (b, 0), (b, 1), (b, 2)\}.$$

2. $(2, a) \in A \times B$; $(a, 2) \notin A \times B$.

3. Examples of ambient sets:

$$(3, -1) \in \mathbb{Z}^2, \quad \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} \in \mathbb{Z}^3, \quad (1, \sqrt{2}) \in \mathbb{R}^2, \quad (4, 7, 9, 0) \in \mathbb{Z}^4.$$

4. $\mathbb{R} \times \{0\}$ is the x -axis.

Scalars and vectors

A vector space is defined relative to a set of **scalars**. In this course the scalar set will be

$$\mathbb{R}.$$

Thus:

- ▶ scalars are real numbers $\alpha, \beta, \dots$;
- ▶ vectors are elements $\mathbf{u}, \mathbf{v}, \dots$ of a set V .

The two types of objects play different roles:

$$\alpha \in \mathbb{R}, \quad \mathbf{v} \in V.$$

Vector space: informal definition

Definition

A **vector space over** $\mathbb{R}$ is a set V whose elements can be

1. added to each other;
2. multiplied by real scalars,

with the usual algebraic behaviour of vector operations.

We write the operations as

$$V \times V \longrightarrow V, \quad (\mathbf{u}, \mathbf{v}) \longmapsto \mathbf{u} + \mathbf{v},$$

$$\mathbb{R} \times V \longrightarrow V, \quad (\alpha, \mathbf{v}) \longmapsto \alpha \mathbf{v}.$$

Vector addition in $\mathbb{R}^n$

For

$$\mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix},$$

vector addition is defined coordinate by coordinate:

$$\boxed{\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{pmatrix} .}$$

Example in $\mathbb{R}^3$:

$$\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 3 \end{pmatrix} .$$

Scalar multiplication in $\mathbb{R}^n$

For $\alpha \in \mathbb{R}$ and

$$\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix},$$

scalar multiplication is

$$\alpha \mathbf{v} = \begin{pmatrix} \alpha v_1 \\ \vdots \\ \alpha v_n \end{pmatrix}.$$

Example:

$$-2 \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 6 \\ -4 \end{pmatrix}.$$

Closure of the operations

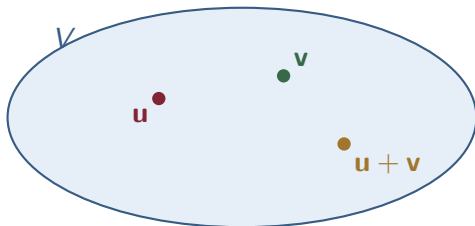
The notation

$$V \times V \longrightarrow V$$

means that if $\mathbf{u}, \mathbf{v} \in V$, then $\mathbf{u} + \mathbf{v}$ is again an element of V . Similarly,

$$\mathbb{R} \times V \longrightarrow V$$

means that if $\alpha \in \mathbb{R}$ and $\mathbf{v} \in V$, then $\alpha\mathbf{v} \in V$.



Standard examples

The fundamental examples for this course are

$$\mathbb{R}, \quad \mathbb{R}^2, \quad \mathbb{R}^3, \quad \mathbb{R}^n.$$

For each of them:

- ▶ vectors are real coordinate lists;
- ▶ addition is coordinatewise;
- ▶ scalar multiplication is coordinatewise.

For example, in $\mathbb{R}^4$,

$$\begin{pmatrix} 1 \\ 2 \\ 0 \\ -1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ -1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 6 \\ 2 \end{pmatrix}.$$

Exercise: operations in $\mathbb{R}^n$

Exercise Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

1. Compute $\mathbf{u} + \mathbf{v}$, $2\mathbf{u}$, $-3\mathbf{v}$ and $2\mathbf{u} - \mathbf{v}$.
2. Find $\mathbf{w} \in \mathbb{R}^3$ such that

$$\mathbf{u} + \mathbf{w} = \mathbf{v}.$$

3. In $\mathbb{R}^4$, compute

$$2 \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 2 \\ 4 \\ -2 \end{pmatrix}.$$

4. Explain why the result of each operation remains in the same ambient space.

Solution: operations in $\mathbb{R}^n$

Solution

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \quad 2\mathbf{u} = \begin{pmatrix} 2 \\ -4 \\ 6 \end{pmatrix}, \quad -3\mathbf{v} = \begin{pmatrix} -6 \\ -3 \\ 3 \end{pmatrix}.$$

$$2\mathbf{u} - \mathbf{v} = \begin{pmatrix} 0 \\ -5 \\ 7 \end{pmatrix}, \quad \mathbf{w} = \mathbf{v} - \mathbf{u} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}.$$

For the $\mathbb{R}^4$ computation,

$$2 \begin{pmatrix} 1 \\ 0 \\ -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 2 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 2 \\ 2 \end{pmatrix}.$$

All results remain in the same ambient space.

Generated vectors and span

Let $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$. Using only vector addition and multiplication by real scalars, we can form vectors of the type

$$a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k, \quad a_1, \dots, a_k \in \mathbb{R}.$$

Definition

The set of all vectors obtainable in this way is the **span** of $\mathbf{v}_1, \dots, \mathbf{v}_k$:

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k : a_i \in \mathbb{R}\}.$$

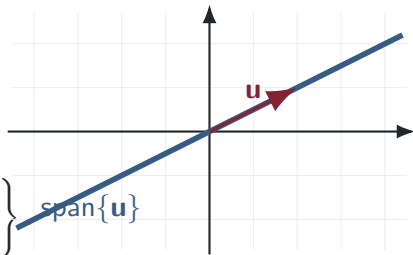
Span of one vector in $\mathbb{R}^2$

Let

$$\mathbf{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Then

$$\text{span}\{\mathbf{u}\} = \left\{ a \begin{pmatrix} 2 \\ 1 \end{pmatrix} : a \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} 2a \\ a \end{pmatrix} : a \in \mathbb{R} \right\}$$



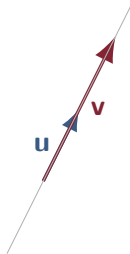
Thus the span is a line through the origin.

Span of two vectors in $\mathbb{R}^2$

Parallel directions

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2\mathbf{u}.$$

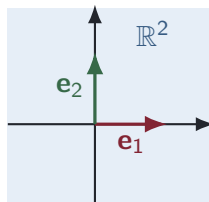
$$\text{span}\{\mathbf{u}, \mathbf{v}\} = \text{span}\{\mathbf{u}\}.$$



Two distinct directions

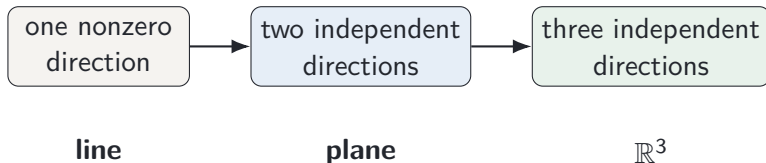
$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$\text{span}\{\mathbf{e}_1, \mathbf{e}_2\} = \mathbb{R}^2.$$



Span in $\mathbb{R}^3$

The geometry depends on the number of nonredundant directions.



Examples:

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\}, \quad \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}, \quad \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Generating sets and redundancy

A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ **generates** V if

$$V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}.$$

A generating set may contain redundant vectors. For example,

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 3\mathbf{e}_1 + 2\mathbf{e}_2.$$

Then

$$\text{span}\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{w}\} = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\} = \mathbb{R}^2.$$

The vector $\mathbf{w}$ does not enlarge the span.

Basis: a nonredundant generating set

Informally, a **basis** of a vector space V is a generating set with no redundant vector.
For $\mathbb{R}^2$,

$$\mathcal{B}_2 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

is the standard basis. For $\mathbb{R}^3$,

$$\mathcal{B}_3 = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

is the standard basis. The precise criterion “spanning and linearly independent” will be stated later in the lecture.

The standard basis of $\mathbb{R}^2$

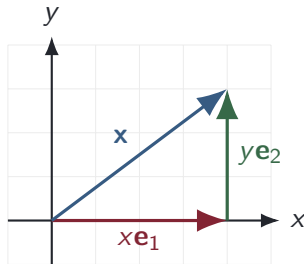
Every vector

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$$

can be written as

$$\mathbf{x} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2.$$

Hence $\mathbf{e}_1, \mathbf{e}_2$ generate $\mathbb{R}^2$.



Dimension

Definition

The **dimension** of a finite-dimensional vector space is the number of vectors in any basis of that space.

Thus

$$\dim(\mathbb{R}) = 1, \quad \dim(\mathbb{R}^2) = 2, \quad \dim(\mathbb{R}^3) = 3, \quad \boxed{\dim(\mathbb{R}^n) = n}.$$

Geometrically:

dim = 1: line

dim = 2: plane

dim = 3: space

Coordinates with respect to a basis

Let

$$\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$$

be a basis of V . Every vector $\mathbf{v} \in V$ can be written using the basis vectors:

$$\mathbf{v} = c_1\mathbf{b}_1 + \dots + c_n\mathbf{b}_n.$$

The coefficients

$$(c_1, \dots, c_n)$$

are the coordinates of $\mathbf{v}$ with respect to $\mathcal{B}$. For the standard basis of $\mathbb{R}^2$,

$$\begin{pmatrix} x \\ y \end{pmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2.$$

Exercise: span, basis and dimension

Exercise

1. Describe geometrically

$$\text{span} \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}.$$

2. Determine whether each set generates $\mathbb{R}^2$:

$$S_1 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \quad S_2 = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix} \right\}.$$

3. Is

$$S_3 = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}$$

a basis of $\mathbb{R}^2$? Give a reason.

4. State $\dim(\mathbb{R}^5)$ and write its standard basis.

Solution: span, basis and dimension

Solution

1. A line through the origin in the direction $(2, 1)$.
2. S_1 generates $\mathbb{R}^2$. The vectors in S_2 are parallel, hence S_2 generates only one line.
3. S_3 generates $\mathbb{R}^2$, but it contains a redundant vector since

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Therefore S_3 is not a basis.

4. $\dim(\mathbb{R}^5) = 5$ and the standard basis is

$$\mathbf{e}_1 = (1, 0, 0, 0, 0), \dots, \mathbf{e}_5 = (0, 0, 0, 0, 1).$$

Linear combination: formal definition

Definition

Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$, a vector of the form

$$a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k, \quad a_1, \dots, a_k \in \mathbb{R},$$

is called a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_k$.

The numbers $a_1, \dots, a_k$ are the coefficients of the linear combination. Consequently,

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{\text{all linear combinations of } \mathbf{v}_1, \dots, \mathbf{v}_k\}.$$

Membership in a span

To determine whether

$$\mathbf{w} \in \text{span}\{\mathbf{u}, \mathbf{v}\},$$

we ask whether there exist scalars $a, b \in \mathbb{R}$ such that

$$a\mathbf{u} + b\mathbf{v} = \mathbf{w}.$$

Example:

$$\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

Coordinate comparison gives

$$\begin{cases} a + 2b = 5, \\ a - b = 1. \end{cases}$$

Thus a span-membership problem leads naturally to a linear system.

Exercise: find the coefficients

Exercise Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

1. Find $a, b \in \mathbb{R}$ such that

$$a\mathbf{u} + b\mathbf{v} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

2. Determine whether $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \text{span}\{\mathbf{u}, \mathbf{v}\}$.

3. In $\mathbb{R}^3$, determine whether there exist a, b such that

$$a \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

Solution: find the coefficients

Solution For the first problem,

$$\begin{cases} a + 2b = 5, \\ a - b = 1, \end{cases} \implies b = \frac{4}{3}, \quad a = \frac{7}{3}.$$

For $(1, 4)$,

$$\begin{cases} a + 2b = 1, \\ a - b = 4, \end{cases}$$

has a solution, so $(1, 4)$ belongs to the span. In $\mathbb{R}^3$ the first two coordinates force $a = 2$, $b = -1$, but then

$$a + b = 1,$$

which matches the third coordinate. Hence the coefficients exist.

Linear dependence

Definition

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly dependent** if there exist scalars $a_1, \dots, a_k$, not all zero, such that

$$a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}.$$

Dependence means that at least one vector is redundant. Example:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$

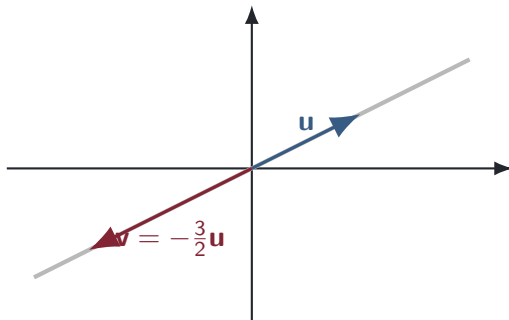
Since

$$2\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0},$$

the vectors are linearly dependent.

Geometric dependence in $\mathbb{R}^2$

Two nonzero vectors in $\mathbb{R}^2$ are linearly dependent exactly when they are parallel.



$$v = \lambda u \implies \lambda u - v = 0.$$

Linear independence

Definition

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if

$$a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k = \mathbf{0}$$

implies

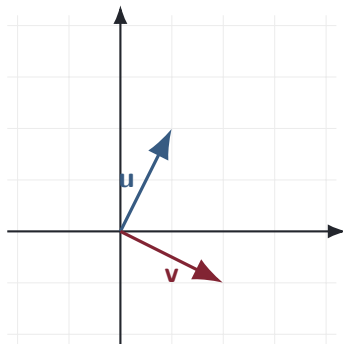
$$a_1 = \dots = a_k = 0.$$

The only representation of the zero vector is the trivial one. In $\mathbb{R}^2$, two nonparallel vectors are linearly independent.

Independent directions in $\mathbb{R}^2$

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

These vectors are not parallel. Hence neither can be obtained by scalar multiplication of the other. They determine two independent directions.



The zero-combination test

To study linear dependence, solve

$$a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k = \mathbf{0}.$$

There are two possibilities:

only the trivial solution
 $a_1 = \cdots = a_k = 0$

linearly independent

at least one nontrivial solution
some $a_i \neq 0$

linearly dependent

Basis: formal criterion

The previous definitions give the standard characterization.

Basis

A set $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis of V if

1. $\mathcal{B}$ spans V ;
2. the vectors in $\mathcal{B}$ are linearly independent.

Therefore a basis is simultaneously

large enough to generate V and small enough to contain no redundancy.

For a finite-dimensional vector space, every basis has the same number of vectors; this number is $\dim V$.

Exercise: dependence and independence

Exercise Classify each set as linearly dependent or linearly independent.

1.

$$\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix} \right\}.$$

2.

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}.$$

3.

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

4.

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Solution: dependence and independence

Solution

1. Dependent, since

$$\begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

2. Independent: the two vectors are not parallel.

3. Dependent, since

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

4. Independent. In

$$a \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0},$$

the first two coordinates imply $a = 0$ and $b = 0$.

Integrated exercise I

Exercise Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}.$$

1. Determine whether $\mathbf{w} \in \text{span}\{\mathbf{u}, \mathbf{v}\}$.
2. Find coefficients a, b if they exist.
3. Are $\mathbf{u}$ and $\mathbf{v}$ linearly independent?
4. Does $\{\mathbf{u}, \mathbf{v}\}$ form a basis of $\mathbb{R}^2$?
5. Express $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ in this basis.

Integrated exercise I: solution

Solution For $a\mathbf{u} + b\mathbf{v} = \mathbf{w}$,

$$\begin{cases} a + 2b = 5, \\ 2a + b = 4, \end{cases} \implies a = 1, b = 2.$$

Hence $\mathbf{w} \in \text{span}\{\mathbf{u}, \mathbf{v}\}$. The vectors are not parallel, so they are linearly independent. Since two independent vectors in $\mathbb{R}^2$ span $\mathbb{R}^2$, they form a basis. For

$$a\mathbf{u} + b\mathbf{v} = \begin{pmatrix} 3 \\ 0 \end{pmatrix},$$

we obtain

$$\begin{cases} a + 2b = 3, \\ 2a + b = 0, \end{cases} \implies a = -1, b = 2.$$

Integrated exercise II

Exercise Consider

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

1. Show directly that $\mathbf{v}_3$ is in the span of $\mathbf{v}_1, \mathbf{v}_2$.
2. Determine whether $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.
3. Describe $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ geometrically.
4. Can these three vectors form a basis of $\mathbb{R}^3$?

Integrated exercise II: solution

Solution Since

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2,$$

we have

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}.$$

The coefficients $(1, 1, -1)$ are not all zero, so the three vectors are linearly dependent. Moreover,

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}.$$

Because $\mathbf{v}_1$ and $\mathbf{v}_2$ are independent, their span is a plane through the origin in $\mathbb{R}^3$. Therefore the three vectors do not form a basis of $\mathbb{R}^3$.

Additional practice I

Exercise

1. Let $A = \{-1, 0, 1\}$ and $B = \{0, 2\}$. Determine $A \times B$ and its cardinality.
2. Write the standard basis of $\mathbb{R}^4$.
3. Describe geometrically

$$\text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}$$

as a subset of $\mathbb{R}^3$.

Additional practice II

Exercise

1. Determine whether

$$\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -4 \\ -2 \end{pmatrix} \right\}$$

is linearly independent.

2. Determine whether

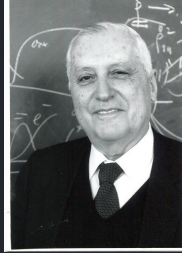
$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

is a basis of $\mathbb{R}^3$. State the dimension of $\mathbb{R}^3$.

3. Find a nonstandard basis of $\mathbb{R}^2$.

Definitions to retain

Cartesian product	$A \times B = \{(a, b) : a \in A, b \in B\}$
Vector space	A set with vector addition and scalar multiplication
Span	All vectors generated from a family by scalar multiplication and addition
Linear combination	$a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k$
Linear independence	The zero combination has only the trivial coefficients
Basis	A spanning, linearly independent family
Dimension	Number of vectors in a basis



“in equilibrium matter is blind, but in far-from-equilibrium conditions it begins to be able to perceive, to take into account, in its way of functioning, differences in the external world.”

Elya Prigogine, Nobel Prize in Chemistry